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Summary

Data Scientist (M.S. in Data Science, Texas A&M) with hands-on experience in Python, R, SQL, and machine learning across research and applied projects in customer analytics, energy markets, and geospatial data. Comfortable taking a project from raw data through modeling to a working dashboard or pipeline, including a recent internship building a document-ingestion pipeline for a RAG-based system in oil and gas operations. Experienced working alongside subject-matter experts outside data science to keep analysis focused on the questions that matter to the business.

Skills

- Languages & Frameworks:** Python, R, SQL, Java, C++, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas
- Machine Learning:** CNNs, transformers, transfer learning, multimodal architectures, monocular depth estimation, embedding extraction, hyperparameter optimization (Optuna)
- Computer Vision & 3D:** OpenCV, Open3D, GLPN, DPT, point cloud processing, Poisson surface reconstruction, depth estimation
- LLMs & NLP:** Hugging Face Transformers, RAG pipelines, Gradio, embedding search
- Data & Engineering:** Apache Spark, PySpark, Hadoop, PostgreSQL, MySQL, Docker, ETL design
- Cloud & Tools:** AWS (EC2, S3, RDS, Lambda), Git, Linux

Projects

Spanish Energy Market Dashboard | Python, Streamlit, scikit-learn, Plotly, Groq API | [Demo](#)

- Built a six-tab Streamlit dashboard analyzing four years of hourly ENTSO-E electricity data (~35K records) covering day-ahead prices, generation mix, and demand for the Spanish power market.
- Built a pipeline joining hourly grid data with multi-city weather data (timezone normalization, deduplication, population-weighted temperature) and modeled the demand-temperature relationship using heating and cooling degree days.
- Trained a next-day demand forecasting model (linear regression with degree-day, calendar, and lagged-load features), reaching roughly 6% MAPE on a 60-day held-out test window.
- Built a day-ahead/spot spread trading-signal and backtesting engine (chronological train/test split, transaction costs, always-long benchmark) across roughly 7,000 simulated trades, then traced part of the apparent edge back to a structural day-ahead/spot bias rather than pure model skill.

Customer Intelligence Platform: Segmentation & BI Dashboard | Python, Streamlit, Plotly | [Demo](#)

- Built a five-page interactive dashboard on a 2,000-customer dataset, letting non-technical users filter by age, income, and gender with live-updating visuals.
- Used KMeans clustering on income, spending, and age to segment customers into four personas (Premium Loyalists, Untapped High-Earners, Aspirational Spenders, Budget Conscious), and wrote up each as a plain-language profile for the marketing team.
- Turned the segmentation into business recommendations, flagging the highest-value segment to retain and the high-earning, low-spending group with the clearest upside.

Air Pollutant Correlation: Uncovering Relationships | Python | [Demo](#)

- Analyzed correlations among multiple air pollutants and environmental variables to identify key drivers of air quality.
- Built linear regression and ensemble models to predict air quality from sensor data ($R^2 = 0.87$), and reviewed feature importance to keep the model interpretable.

DepthForge: Monocular 3D Reconstruction | PyTorch, Open3D, Hugging Face Spaces | [Demo](#)

- Built a depth-to-3D pipeline: a single RGB photo goes through a transformer depth model (GLPN-NYU or Intel DPT-Large), then gets back-projected into a point cloud and reconstructed into a mesh via Poisson surface reconstruction.
- Added mesh quality checks (manifold edge/vertex tests, watertightness, surface area, volume) to judge whether the reconstructed geometry was actually usable, not just plausible-looking.
- Compared GLPN vs. DPT across indoor and outdoor scenes on outlier percentage and topology metrics, and deployed the pipeline on Hugging Face Spaces with a responsible-use consent gate, EXIF privacy checks, and PLY/OBJ/STL export.

Experience

AI Intern, OPX AI, Houston, Texas Jan 2026 – Apr 2026

- Built a PDF ingestion pipeline for a safety-critical RAG system used in oil and gas operations, turning technical and regulatory documents into structured, traceable chunks so LLM responses could be grounded and checked back against the source.
- Refactored the pipeline into a modular Python package with typed dataclasses, parallel UUID-based ingestion to avoid collisions, atomic writes, structured logging, and a full unit test suite to meet the reliability bar for a regulated production environment.

Research Data Scientist, Texas A&M University, College Station, Texas Aug 2024 - Jan 2026

- Designed clustering pipelines (DBSCAN, hierarchical, sequential, and graph-based methods) to analyze large-scale GOES GLM satellite lightning data, helping the team identify and characterize extreme lightning events.
- Built interactive geospatial dashboards (Python, Leaflet, GeoPandas) so researchers could explore lightning frequency, altitude, and spatial clusters directly, cutting down on one-off scripts for routine exploratory analysis.
- Worked closely with atmospheric scientists to pressure-test assumptions and refine evaluation metrics, keeping the analysis focused on questions that mattered to the research.

Research Assistant, SGL Lab, Texas A&M University, College Station, Texas Feb 2024 – Apr 2024

- Cleaned and prepared global supply-chain datasets, including trade flow and commodity data, using Python, R, and SQL.
- Built a graph-based dependency model of a 3-node semiconductor supply chain to study systemic risk propagation, applying probabilistic and statistical methods to assess uncertainty.

Machine Learning Engineer, IIT Bombay, Mumbai, India Jan 2022 – Nov 2022

- Curated a labeled dataset of 26 Indian Sign Language gestures from video footage (Python, OpenCV), extracting per-frame features for reproducible model training.
- Trained and compared CNN architectures, including VGG16 with transfer learning, on accuracy, loss curves, and per-class metrics to select the best model for sign language recognition; this work was recognized with the TIH-IoT CHANAKYA Fellowship.

Education

Texas A&M University, MS in Data Science GPA: 3.9/4 Aug 2023 – Dec 2024

Anna University, B. E in Computer Science and Engineering GPA: 8.78/10 Jun 2019 – May 2023

Publication

"Secure Smart Cabin Using Optimized Arduino GSM Interface" Springer [Link](#)